



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	OPTICS				
Type of Course	Core in Major				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamental understanding of basic physics principles, including optics, electromagnetic waves, and mathematical concepts such as calculus and trigonometry.				
Course Summary	The course offers an in-depth study of light phenomena, covering polarization effects, diffraction phenomena, and their applications in optical systems and technologies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define the basic principles of optics.	Remember	Definitions and basic concepts	Quizzes

CO2	Analyse optical phenomena using Fermat's Principle, such as reflection and refraction.	Analyse	Optical phenomena and causes	Research papers, case studies
CO3	Apply the principles of optics to design optical systems.	Apply	Application of principles	Laboratory experiments, projects
CO4	Analyse optical phenomena interference	Analyse	Optical phenomena and causes	Research papers, case studies
CO5	Apply diffraction principles to analyze patterns produced by various apertures and obstacles.	Apply	Application of principles	Laboratory experiments, simulations
CO6	Apply polarization concepts to analyze optical phenomena.	Apply	Application of principles	Laboratory experiments, simulations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	FERMAT'S PRINCIPLE		8	15
	1	Laws of reflection and refraction from Fermat's Principle	2	
	2	Refraction and reflection at a single Spherical surface	2	
	3	The thin lens , Principal foci and focal length	2	

	4	The Newton formula, lateral magnification	2	
	Sections 3.1, 3.2, 4.1 – 4.7 of Book 1			
II	INTERFERENCE		16	25
	5	Superpositions of two sinusoidal waves	1	
	6	Interference division of wavefront introduction	1	
	7	Interference of light waves	3	
	8	Fresnel's two mirror and Fresnel's Biprism	2	
	9	Interference with white light , Lloyd's mirror , Phase change on reflection	3	
	10	Interference by division of amplitude -Non reflecting films	3	
	11	Colours of thin films, Newton's rings, Michelson interferometer	3	
	Sections 13.5, 14.1, 14.3 – 14.12 , 15.1 – 15.4, 15.8 – 15.11 of Book1			
III	DIFFRACTION		10	15
	12	Single- Slit Fraunhofer diffraction pattern	2	
	13	Two Slit Fraunhofer diffraction pattern	2	
	14	N Slit Fraunhofer diffraction pattern and Grating	2	
	15	Fresnel diffraction – Zone plate	2	
	16	Diffraction by straight edge	2	
	Sections 18.1, 18.2, 18.6 – 18.8, 20.1 – 20.3, 20.6 of Book 1			
IV	POLARISATION		11	15
	17	Polarisation Introduction	1	
	18	Production of linearly polarised light	2	

	19	Effects of polariser and analyser	1	
	20	Double refraction -Huygens' explanation	2	
	21	Wave plates	2	
	22	Production and analysis of different polarised light	3	
	Sections 20.1 – 20.4, 20.5, 20.6.2 – 20.6.3, 20.8.3, 20.9.1, 20.17 – 20.20 of Book 2			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Determine the refractive index of (a) given liquid and (b) the material of a lens, by forming a liquid lens. - Through this experiment the students are expected to get the concepts of image formation, combination of lenses and radius of curvature of the surface of lens. - Determine the radius of curvature of the lens by Boy's method and hence calculate the refractive indices.		
	2	Determine the focal length of the combination of two lenses separated by a distance. - Determine the focal lengths, f_1 and f_2 of the two lenses using an illuminated cross-slit screen holder, nodal slide(for placing the lenses) and plane mirror arrangement. - Place the two lenses separated by a distance d, determine the focal length, F of the combination and verify the relation $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$. - The combination of the lenses in the eyepiece of the spectrometer/ travelling microscope may be used for the study. https://www.youtube.com/watch?v=IOIEEtyNPBg https://www.youtube.com/watch?v=tNo4Ipk74SU		
	3	Determination of the dispersive power of a solid prism using a spectrometer. Find the angle of the prism and the angle of minimum deviation for prominent lines of the mercury spectrum using a spectrometer.		

	Calculate the refractive indices corresponding to the colors and find the dispersive power of the material of the prism for two pairs of wavelengths.		
4	Refractive indices of quartz prism using spectrometer. - Determine the refractive indices of quartz for the ordinary and extraordinary rays of a sodium vapour lamp by arranging the quartz prism at minimum deviation position in the spectrometer. - Verify the polarizations of the ordinary and extraordinary rays using a polaroid.		
5	Determination of wavelengths of mercury spectrum using diffraction grating and spectrometer. - Arrange the grating at normal incidence. - Standardize the grating using the green line of mercury and then find the wavelengths of other prominent lines of the spectrum.		
6	Newton's rings-determination of the wavelength of sodium light - Form of Newton's rings in the air-film in between a plano-convex lens and a glass plate using sodium-source. - Determine the radius of curvature by Boy's method and determine the wavelength of the source. - Optional: In experiment 5 and 6, record a short video of the interference pattern, calibrate the video using scale marked on the glass plate, analyse the video using Tracker tool. From the intensity profile get the locations of the dark rings and calculate the wavelength of the source/thickness of the sample https://physlets.org/tracker/ https://www.youtube.com/watch?v=UCCPkIpUQEw		
7	Air wedge-determination of the radius of a thin wire/human hair/thin foil. - Form interference fringes using sodium-source, in the air-film in between wedge formed by placing the given sample between the glass plates. - Measure the positions of the successive dark bands using a travelling microscope and determine the angle of the wedge and thickness of the sample given.		
8	Single slit diffraction using laser - Determination of slit width. - The laser light diffracted from the narrow slit is allowed to fall on a screen and record the maxima or minima points in a paper. - From the width of the central maxima or the position of minimum intensity points, calculate the slit width. - Verify the slit width using a traveling microscope. - Wavelength of laser can be found using diffraction grating of known N.		

9	Analysis of the diffraction patterns using Tracker tool. - Project the diffracted laser light from a narrow wire/single slit/double slit/small rectangular/circular aperture into a screen/wall and record a clear short video/image of the diffraction pattern. - Analyse the video/image using Tracker tool - calibrate using the scale marked on the screen/wall and plot the intensity profile. From the location of the maxima/minima, determine the dimension of the scattering source. - The size of the scattering source may be verified using a microscope or by analysing the HD image of the source with an image editing software. https://physlets.org/tracker/. https://www.youtube.com/watch?v=UCCPkIpUQEw		
10	Study the specific rotation of the sugar solution using a polarimeter. - Determine the specific rotation corresponding to different concentrations of the sugar dissolved in water. - Draw a graph between rotation and concentrations and verify the linear relationship.		
11	Verification of Malus's law using polarizer, analyzer and photo detector - Unpolarized light is allowed to pass through a polarizer and is observed through an analyzer. - Vary the angle between the axes of polarizer and analyzer and measure the intensity of the light (current output of the photodetector). - Plot $\theta - I$ and $\cos^2 \theta - I$ graphs and verify Malus' law.		
12	Spectrometer-Determination of the Cauchy's constants of the given prism - Find the angle of the prism, the minimum deviation angles of the prominent lines of the mercury spectrum and hence calculate the refractive indices for the colors. - Determine A and B from the $\mu - \frac{1}{\lambda^2}$ graph.		

Books and References:

- Optics by Ajoy Ghatak; 6th Edition (Book 1)
- A Text Book of Optics by N. Subrahmanyam, Brij Lal and M.N Avadhanulu; 2018 Revised Edition (Book 2)
- Optics by Eugene Hecht
- Introduction to Modern Optics by Grant R. Fowles

5. Introduction to optics by Frank L. Pedrotti, Leno M. Pedrotti
6. Fundamentals of Optics by Jenkins F

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO 5	PSO 6	PO 1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2	0	0	0	2	0	2	1	0	0	1	1	1
CO 2	2	2	0	2	2	0	2	1	1	0	1	1	1
CO 3	2	1	0	3	2	1	2	1	1	1	1	1	1
CO 4	2	2	0	2	2	0	2	1	1	0	1	1	1
CO 5	2	1	0	2	2	1	2	1	1	1	1	1	1
CO 6	2	1	0	2	2	1	2	1	1	1	1	1	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	